

6.2

máme bázi $B = \left(\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right)$ prostoru V .

$$[(4, 1, 2, 5)^T]_B = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad [(2, 2, -2, -2)^T]_B = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$

Pak $[f]_B^B$ zobrazí $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ na $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ a $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ na $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$,
tedy $[f]_B^B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$

Nyní nalezneme $[f]_{k_4}^{k_4} = [id]_{k_4}^B [f]_B^B [id]_B^{k_4}$

$$[id]_{k_4}^B = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \\ -1 & 2 \end{pmatrix} \quad [id]_B^{k_4} = [id]_{k_4}^B^{-1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\rightarrow \text{tímě } \left(\begin{array}{cc|cccc} 2 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \end{array} \right)$$

$$\text{tedy } [f]_{k_4}^{k_4} = \begin{pmatrix} 0 & 4 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 5 & -2 & 0 \end{pmatrix}$$

Nalezneme vlastní čísla této matice. $\det([f]_{k_4}^{k_4} - \lambda I_4) = x^2(x-2)(3+x)$

Nalezneme lineární polynom a příslušné vlastní vektory jako $\ker(f - \lambda I)$

$$\lambda_1 = -3 \quad M_{-3} = \text{Span} \{ (0, -1, 2, 3)^T \}$$

$$\lambda_2 = 2 \quad M_2 = \text{Span} \{ (5, 2, 1, 4)^T \}$$

$$\lambda_3 = 0 \quad M_0 = \text{Span} \{ (0, 0, 0, 1)^T, (1, 0, 0, 0)^T \}$$

Ukážeme bázi C a těchto vlastních vektorů $C = \left(\begin{pmatrix} 0 \\ -1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right)$

$$\text{Pak } [id]_{k_4}^C = \begin{pmatrix} 0 & 5 & 0 & 1 \\ -1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 4 & 1 & 0 \end{pmatrix}$$

$$[f]_C^C = \begin{pmatrix} -3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$[id]_{k_4}^C = [id]_C^{k_4} [id]_{k_4}^{k_4} = \begin{pmatrix} 0 & 5 & 0 & 1 \\ -1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 4 & 1 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 5 & -10 & 5 \\ 0 & -5 & 10 & -5 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$[f]_{k_4}^{k_4^m} = [id]_{k_4}^c [f]_c^{k_4^m} = \frac{1}{5} \begin{pmatrix} 10 \cdot 2^m & 5 \cdot 2^m & 0 & 0 \\ (-3)^m + 4 \cdot 2^m & -4 \cdot 3^m + 2 \cdot 2^m & 0 & 0 \\ -2(-3)^m + 2 \cdot 2^m & 4(-3)^m + 2^m & 0 & 0 \\ -8(-3)^m + 8 \cdot 2^m & 6(-3)^m + 4 \cdot 2^m & 0 & 0 \end{pmatrix}$$

Pak $f^m((5, 1, 3, 7)^T) = [f]_{k_4}^{k_4^m} \cdot (5, 1, 3, 7)^T$

$$a [id]_c^{k_4} = \frac{1}{5} \begin{pmatrix} 0 & -1 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & -5 & -10 & 5 \\ 5 & -10 & -5 & 0 \end{pmatrix}^{-1} = [id]_{k_4}^c$$

★ Pak maine:

$$[f^m]_{k_4}^{k_4} = [id]_{k_4}^c [f^m]_c^c [id]_c^{k_4} = () \quad \text{nedopisat'emo}$$

$$a f^m((5, 1, 3, 7)^T) = [f^m]_{k_4}^{k_4} \cdot \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix}$$